

Shuvam Thakur

SOFTWARE ENGINEER

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PROFILE SUMMARY

Software engineer with a proven track record in a global investment-banking technology team, specializing in full-stack development, distributed systems, and large-scale data engineering. Experienced in managing platform systems that calculate counterparty exposure and credit limits across global trading portfolios. Pursuing advanced postgraduate studies to deepen expertise in Artificial Intelligence architectures, machine learning models, and predictive analytics.

EDUCATION

University of Technology Sydney (UTS)

Master of Artificial Intelligence | July 2026 – Present

Delhi Technological University (DTU)

Bachelor of Technology in Computer Engineering (GPA: 8.23) | 2019 – 2023

PROFESSIONAL EXPERIENCE

Macquarie Global Services • Senior Associate (L2 Software Engineer)

April 2025 – June 2026

Engineer on the firm's strategic credit-risk platform, owning applications that calculate counterparty exposure and credit limits across the global trading book.

CORE DELIVERY & BUSINESS IMPACT

- **Technical SME:** Recognized as the subject-matter expert for platform calculations and data validations; served as primary engineering contact for credit-risk business users.
- **Production Stability:** Resolved 245 critical tickets (defects, stories, incidents) across the credit-risk platform, the exposure database, and downstream batch pipelines.
- **Division Upgrade:** Led the technical execution of a multi-month core data-model expansion, orchestrating 8 deployment bundles across 100+ database tables and dozens of stored procedures.
- **Infrastructure Modernization:** Contributed to a global Sybase-to-Postgres replication migration and led the division-wide PKI certificate refresh across 12+ services.
- **Technical Documentation:** Authored 36+ internal Confluence architecture pages and runbooks read by engineering teams.
- **Reference:** Karan Juneja (Manager and Credit Risk Tech Lead, Macquarie Group) | +61 412 036 110

TECHNICAL DEEP DIVES

- **Incident Resolution:** Traced a calculation pipeline bug through a 17-step state inconsistency between adjacent handlers, spanning five microservices and the database.
- **Data Warehouse Debugging:** Root-caused a missing rating defect in the scoring application where Hibernate dirty-checking was silently overwriting a final-rating column for ~500 records.
- **Batch Ingestion Fixes:** Resolved a duplicate data incident on a daily batch by tracing an upstream API failing to push to a downstream allocation table, producing an automated safeguard script.
- **Engineering Contribution:** Contributed to 6+ Spring Boot microservices, logged 720+ commits, and maintained an Angular microfrontend ecosystem comprising 12 micro-frontends.

Macquarie Global Services • Associate Engineer

November 2023 – March 2025

- **Backend Development:** Built and deployed enterprise-grade Spring Boot microservices to RESTful API standards, delivering validation refactors and batch job performance fixes.
- **Full-Stack SDE:** Developed responsive Angular dashboards and enforced 90%+ JUnit code coverage using JaCoCo gating.
- **DevOps & Security:** Owned Bamboo CI/CD workflows and managed application security (Fortify, Black Duck) and SSL/p12 certificate refresh via Cyberark.

Macquarie Global Services • Full-Stack SDE Intern

January 2023 – October 2023

- Collaborated with senior engineers to implement RESTful Spring Boot microservices for risk-calculation modules.
- Developed responsive Angular frontend dashboards, ensuring real-time risk data visualization.
- Configured CI/CD pipelines using Bamboo and actively participated in Agile ceremonies and code reviews.

PROFESSIONAL EXPERIENCE (CONTINUED)

Airtel Payments Bank • Backend Engineering Intern

June 2022 – August 2022

- Engineered and optimized Java Spring Boot backend microservices contributing to secure payment-processing.
- Developed responsive customer-facing banking portal interfaces utilizing Angular 8 templates.
- Integrated third-party and internal RESTful APIs, ensuring real-time consistency and high availability.

LIS • Data Engineering Intern

August 2021 – November 2021

- Architected and deployed automated Python web-scraping scripts to harvest raw text data from unstructured sources.
- Curated, cleaned, and pre-processed large-scale text datasets for Natural Language Processing (NLP) model training.

Rato Bangala Kitab • Social Media & Graphics Design Intern

June 2018 – January 2019

- Designed brand-aligned graphical content, infographics, and posters, managing regular website updates.
- Directed multimedia production and post-production cinematography editing for the RBK YouTube recital channel.

Sofos Softwares Pvt. Ltd. • Web Designer Intern

December 2017 – January 2018

- Assisted in web design and frontend development utilizing HTML, CSS, and JavaScript for interactive client pages.
- Collaborated with core developers to troubleshoot UI/UX issues, ensuring cross-browser compatibility.

SELECTED PROJECTS

Early Detection of Alzheimer's Disease via Speech Analysis [Deep Learning / NLP](#)

- **Architecture Design:** Engineered a hybrid CNN + BiLSTM (2 layers) network with max-pooling to extract linguistic features from speech transcripts.
- **Benchmark Analysis:** Achieved 92% accuracy and F1-macro of 0.92 benchmarking custom models against pre-trained transformer blocks (AlBERT) for early-stage AD in Hindi.

Multimodal Transformer Adaptation [Computer Vision / Social Computing](#)

- **Framework Customization:** Designed task-specific multimodal classifiers utilizing CLIP-ViT backbones, developing a Co-Attention Ensemble model for Hate Speech detection (F1 score: 0.7929).

TECHNICAL SKILLS

Languages & Frameworks: Java (Spring Boot, JDBC, Hibernate), Python (Airflow, Pandas), TypeScript, JavaScript (Angular UIs), HTML/CSS, SQL (T-SQL, PL/SQL), Bash, Scala (Spark)

Backend & Distributed: RESTful APIs, Drools rules, Apache Spark, Apache Kafka, Solace JMS, Apache Airflow, Quartz

Databases & Cloud: MS SQL Server, Sybase, Oracle, MongoDB, MySQL, Postgres, SQLite, Presto/Trino, AWS Athena, AWS (RDS, S3, EKS, EMR, IAM), Docker, Bamboo CI/CD, Git, Fortify

ML & Observability: TensorFlow, Keras, PyTorch, HuggingFace, scikit-learn, NumPy, NLP, Deep Learning, Computer Vision, Grafana, Splunk, Postman, JUnit + JaCoCo

ACHIEVEMENTS & AWARDS

- **Google Cloud Certified:** Data Science & ML Track
- **Duke of Edinburgh:** Bronze Award Recipient (2017)
- **Gold Medalist:** Mahasangram Debate Competition (2017)
- **Global Rank 6:** International Mars Hackathon (2020)
- **Second Runner-Up:** Indian Rover Challenge (2019)